

Supplementary Material for “Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose”

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Submission Id: 2419

Abstract

This supplement supports the main paper with appendices on multiplicity control and the clustered interval on the weak edge (A–B), threshold sensitivity (C), the theory assumptions behind the percolation form and the reproduction number (D), the recurrent case (E), the calibration, scope and failure modes of the compromise judge together with the adjudication of a labelled subset (F), utilities and estimator validation (G), the practical implications and the full limitations (H–I), the depth-profile table (J), ethical and artifact statements (K), the disclosure of AI-assisted technologies (L), and replication details (M). Table and appendix numbers here are independent of the main paper; references to the main paper are given by section number in words.

Keywords

Prompt injection, LLM agent networks, measurement validity, reproducibility

ACM Reference Format:

Anonymous Author(s). 2027. Supplementary Material for “Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose”. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 9 pages.

Appendix A. Multiplicity Control for the Composition Test

Table 1 applies a Benjamini–Hochberg correction to the composition tests of Section 5.9 of the main paper at the common $n = 40$ grid. Only the Llama cell rejects H_0 ; it survives the correction at $q = 0.05$, while the four remaining non-degenerate cells—including DeepSeek, whose fixed-artefact deviation is a protocol artefact that disappears under the fresh protocol (Section 5.3 of the main paper)—stay non-significant. The correction inherits the protocol mix of the cross-model table: the Llama and DeepSeek rows were re-run with fresh artefacts while the qwen, Nova and Claude rows predate that policy (Table 1 of the main paper), so the five p -values are not all computed under identical artefact sampling and a fresh-artefact replicate of the latter three would be the clean version of this table. The Claude cell is degenerate ($ASR = \prod_i \hat{s}_i = 0$, $sd(\Delta) = 0$) and is excluded from the correction. At the higher $n = 200$ resolution the Nova cell also crosses into rejection (Section 5.9 of the main paper);

we keep the correction at the common $n = 40$ grid so the five cells are compared at equal power.

Table 1: Benjamini–Hochberg over the $m = 5$ non-degenerate chain cells (fresh-artefact protocol). p -values are the bootstrap two-sided values from Section 5.9 of the main paper. Only the Llama rejection survives.

rank	cell	p	BH thr. ($q=0.05$)
1	Llama · none	0.0002	0.010
2	Nova · none	0.0840	0.020
3	qwen · none	0.3165	0.030
4	DeepSeek · none (fresh)	0.6300	0.040
5	qwen · paraphrase	0.7775	0.050

Appendix B. Family-Wise Control Extended to the Remaining Families

Appendix A accounted for multiplicity among the five composition cells; the topology, content-form, obfuscation and loop comparisons that the main paper reports as outcomes were previously labelled exploratory without any family-wise accounting. We now extend the same Benjamini–Hochberg control to every null among those families that is testable from the stored artefact, without any new model call (scripts/familywise_extended.py).

Table 2 lists the $m = 12$ tests, the exact p -values (Fisher’s exact test on the observed counts; exact two-sided binomial against the point prediction for the loop), the within-family BH threshold at $q = 0.05$, and the verdict. Every family keeps at least one surviving rejection, and the Claude floor comparison honestly fails (Section D shows there is almost nothing to defend). Three findings deserve to be stated explicitly:

- (1) **The star–tree ASR gap is not significant.** At $n = 7$ (Llama, 40 natural runs per cell) the tree reaches its target in 0.800 of runs versus 0.725 for the star; the difference is $p = 0.60$, within the noise the caption already flags. What is supported—and survives the correction—is the chain–tree gap (0.450 vs. 0.800, $p = 0.0024$) and the chain–star gap (0.450 vs. 0.725, $p = 0.022$). We therefore read the main paper’s “most permissive” as a *numerical* ordering only, and the star–tree reversal with scale rests on the $n = 10$ and DeepSeek replicates (Section P), not on the $n = 7$ gap alone.
- (2) **The content-form effect survives anything we can correct for.** On Llama’s middle edge the same target survives 29/30 embedded versus 2/30 canonical ($p = 2.3 \times 10^{-13}$).



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- (3) **The loop falsifications all survive.** The exact finite chain (Section F) is rejected in every model against the observed last-round manager rate ($p \leq 1.6 \times 10^{-4}$ even for the marginal DeepSeek cell).

A pooled, claim-level BH over all 22 tests that yield a p -value anywhere in the paper (5 composition + 5 depth + 3 topology + 1 content form + 5 obfuscation + 3 loop) leaves the same fourteen rejections as the within-family passes: the composition family’s Llama cell, the depth trends of qwen and both Llama replicates, the two topology gaps named above, the content-form effect, the four non-floor obfuscation tests, and the three loop tests. Nothing that is claimed as significant in the main text is an artefact of unchecked multiplicity, and nothing that fails here is claimed there. The R_0 point comparisons (0.831 vs. 0.821 and 0.420) remain descriptive because, as the topology caption states, their numerator and denominator are not recoverable from the per-cell files we archived, so a test on them would not be reproducible from the release.

Table 2: Exact tests across the formerly uncorrected families, with within-family Benjamini–Hochberg control at $q = 0.05$. Counts are read from the archived per-cell files; no model was called. ✓ = survives the family threshold; × = fails.

family	comparison	counts	exact p	verdict
topology	chain vs. tree	18/40 vs. 32/40	0.0024	✓
topology	chain vs. star	18/40 vs. 29/40	0.0225	✓
topology	star vs. tree	29/40 vs. 32/40	0.60	×
content form	embedded vs. canonical (Llama a1→a2)	29/30 vs. 2/30	2.3×10^{-13}	✓
obfuscation	qwen: redaction removes plain	15/16 → 0/16	5.7×10^{-8}	✓
obfuscation	qwen: split bypasses redaction	16/16 vs. 0/16	3.3×10^{-9}	✓
obfuscation	DeepSeek: redaction removes plain	8/8 → 0/8	1.6×10^{-4}	✓
obfuscation	DeepSeek: split bypasses redaction	8/8 vs. 0/8	1.6×10^{-4}	✓
obfuscation	Claude: redaction removes plain	3/30 → 0/30	0.24	×
loop	qwen: observed vs. prediction	26/40 vs. 0.048	5.0×10^{-25}	✓
loop	Llama: observed vs. prediction	28/40 vs. 0.989	1.3×10^{-14}	✓
loop	DeepSeek: observed vs. prediction	34/60 vs. 0.328	1.6×10^{-4}	✓

Appendix C. Context-Clustered Interval on the Weak Edge

The main paper’s power and multiplicity paragraph reports that a homogeneity test rejects exchangeability across benign task contexts on the weak Llama edge. Because that violates the i.i.d. assumption behind Wilson intervals, we recompute the edge rate with a cluster bootstrap that resamples *repeats* (seed replications), not individual trials. The test is the Pearson statistic on the per-context counts (18/56, 3/56, 13/48 at $T = 0.7$, so $n = 160$ across three contexts): $\chi^2 = 13.40$ on $df = 2$, with the tail evaluated by the Wilson–Hilferty approximation used throughout the code, $p = 0.0014$ (the exact χ^2_2 tail gives 0.0012; the difference does not affect any conclusion). At a fixed temperature ($T = 0.7$, six repeats of twenty trials) the two intervals nearly coincide (Table 3): the design-effect widening is 0.99×, so within a temperature the Wilson interval is well calibrated. The overdispersion $\phi = 2.93$ is a *cross-temperature* artefact—temperature is a deliberate systematic factor, not trial noise—which is why we compute ϕ within a temperature.

Table 3: Weak edge a1 → a2 (Llama, $T = 0.7$, $n = 120$): i.i.d. Wilson versus a repeat-clustered bootstrap (10^4 resamples).

interval	$\hat{s}$	95% CI	width
Wilson (i.i.d.)	0.242	[0.174, 0.326]	0.152
cluster bootstrap (by repeat)	0.242	[0.167, 0.317]	0.150

Appendix D. Threshold Sensitivity of the Floor Effect

A referee may ask whether the “resistant” verdict on the frontier model is an artefact of the compromise threshold $\tau_{ASV} = 0.9$. We re-score the single-hop obfuscation cells (Claude, $n = 30$ per cell) at $\tau \in \{0.7, 0.8, 0.9, 1.0\}$ directly from the stored ASV scores—a deterministic re-thresholding, no new model calls. The compromise rate is *invariant* across τ in every cell (Table 4): the no-defence baseline stays at 0.17 and redaction at 0.00 regardless of the threshold. Note that 0.17 is the *single-hop* no-defence rate for this model and cell; the end-to-end chain rate quoted in the main paper’s headroom discussion is 0.10, a different quantity with a different denominator, and the two are not interchangeable. The reason is that the ASV distribution is bimodal—of 180 trials, 158 score 0.0 and 14 score 1.0; the only eight intermediate trials all score ≤ 0.667 , i.e. below the lowest threshold tested. The floor effect is therefore a property of the model, not of where the judge’s cut is placed.

Table 4: Single-hop compromise rate versus τ_{ASV} (Claude, $n = 30$ per cell), re-scored offline from stored ASV. Rates are threshold-invariant.

defence	form	$\tau=0.7$	0.8	0.9	1.0
none	plain	0.17	0.17	0.17	0.17
none	spaced	0.17	0.17	0.17	0.17
none	split-word	0.13	0.13	0.13	0.13
redact	plain	0.00	0.00	0.00	0.00
redact	spaced	0.00	0.00	0.00	0.00
redact	split-word	0.00	0.00	0.00	0.00

Appendix E. Theory Assumptions: Independence, Correlation, and R_0

This appendix clarifies the theoretical assumptions invoked in Section 3.8 of the main paper.

Product-of-complements and correlation. The percolation recurrence

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u])$$

assumes *edge activations* (and the corresponding parent-reach events relevant to the product form) are conditionally independent given the graph structure. When multiple parents $u_1, \dots, u_k$ of v share a common ancestor, the events $\{\Pr[u_i]\}$ may be correlated, so $\Pr[\bigwedge_i (\text{not reached via } u_i)] \neq \prod_i (1 - s_{u_i \rightarrow v} \Pr[u_i])$ in general. For the specific topologies evaluated in this work (chains, trees, star, and the small manager/worker loop), we do not rely on that product form across shared ancestors; where we use it, we treat it as the marginal approximation under the stated conditions or restrict

claims to graph classes where the independence holds. For arbitrary DAGs with shared ancestors, a path-based (inclusion–exclusion) formulation or explicit correlation modelling is required.

R₀ definition and scope. For strictly acyclic (feed-forward) graphs, the standard linearized next-generation matrix M with $M_{uv} = s_{u \rightarrow v}$ is strictly triangular in a topological ordering, hence $\rho(M) = 0$. This is a structural property of acyclicity, not an empirical safety guarantee. We therefore report R_0 as a *descriptive, finite-horizon* reproduction statistic: specifically, we define R_0 as the mean number of newly compromised neighbours produced per compromised agent instance over the observed finite horizon (generation convention as stated in Section 4.2 of the main paper), with its denominator clearly specified. We do not interpret $\rho(M) < 1$ as a threshold for safety on finite acyclic graphs; reachability is given by the percolation (Eq. 3 in the main paper), not by R_0 alone.

Parallel paths and defence placement. Hardening a single node is not equally effective when there exist multiple s – t paths: effectiveness depends on whether that node lies on all paths (a cut). For unique-path graphs the effect is path-specific; we do not generalise equal effectiveness to multi-path DAGs. Placement claims are limited to the graph classes we evaluate.

Transportability and product composition. The chain identity $\text{ASR} = \prod_i s_i^{\text{nat}}$ is algebraic given the Markov/chain structure. However, composing *isolated* per-edge estimates s_i^{ctrl} requires that $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$ (transportability). We do not assume this holds in general; we test it empirically and use the depth-error slope as a diagnostic (Section 5.2 of the main paper).

Appendix F. Recurrence Makes the Threshold Non-Vacuous

This appendix gives the full treatment of the recurrent (cyclic) case summarised in Section 5.8 of the main paper. We test a pattern real frameworks use: a manager broadcasting to workers that report back. (The complementary test of the percolation recurrence on a reconvergent *acyclic* graph, where it can fail, is Appendix O.)

For a manager with w reporting workers, the infection matrix has $M[m \rightarrow w_j] = a_j$ and $M[w_j \rightarrow m] = c_j$, and its characteristic polynomial is $\lambda^{w-1}(\lambda^2 - \sum_j a_j c_j)$. Hence

$$\rho(M) = \sqrt{\sum_j a_j c_j} \quad (1)$$

for every w (checked numerically against the eigenvalues to machine precision on 320 random configurations). Two consequences follow. A feed-forward reading sets every $c_j = 0$, so $\rho = 0$ however strong the edges are—the vacuity of Section 3.8 of the main paper again, now with the feedback edges removed. And the system is supercritical exactly when $\sum_j a_j c_j > 1$; for symmetric edges $a_j = c_j = s$ this is $w s^2 > 1$. That is a *design rule* rather than a diagnostic: at $s = 0.5$ a manager needs five reporting workers to sustain an endemic injection, at $s = 0.7$ three, at $s = 0.9$ two.

Applied to measured survivals, Eq. (1) makes opposite predictions, which is the test a criterion has to pass to be worth using (Table 5). On Llama the sum is 1.800 and on DeepSeek 1.018, so both frontier loops should be supercritical; on qwen2.5:7b it is 0.578, so the same loop should decay.

Table 5: One criterion, three models, opposite predictions. $\sum_j a_j c_j$ uses the measured per-hop survivals of the same three-agent recurrent pattern; the feed-forward column is the same survivals with the two feedback edges removed.

model	$\sum_j a_j c_j$	ρ rec.	ρ fwd.	prediction
Llama 3.3 70B	1.800	1.342	0.000	supercritical
DeepSeek V3.2	1.018	1.009	0.000	supercritical (marginal)
qwen2.5:7b	0.578	0.760	0.000	subcritical

The measured dynamics follow in *direction*: on Llama the recurrent manager is re-compromised by its own workers’ reports in 0.725 [0.572, 0.839] of runs and does not die out—at round five it is still compromised in 0.700 of runs with the workers near saturation (0.925, 0.975); extending the loop to eight rounds leaves it endemic (manager 0.583, workers 0.900). DeepSeek, whose $\sum_j a_j c_j = 1.018$ sits just above criticality, behaves the same way—the manager settles at 0.567 with workers at 0.767 and 0.700 rather than decaying—so the recurrent verdict is not a single-model artefact. On qwen the same loop decays: the manager’s re-compromise rate falls from 0.775 [0.625, 0.877] to 0.650 and the workers plateau (0.475, 0.525).

The exact finite chain does not reproduce those rates. The mean-field comparison above says only which side of criticality a loop is on, and we do not read it as a model of the per-round process. Building the exact finite chain instead—with w workers, the manager compromised by at least one returning report, and workers able to be compromised only through the manager—gives $m_{k+1} = 1 - \prod_j (1 - a_j c_j m_k)$, which is strictly below one at every round even when $\sum_j a_j c_j > 1$. Table 6 compares that prediction with the observed manager rate round by round.

Table 6: Exact finite-state chain versus the observed manager compromise rate per round. Prediction uses the measured a_j (worker compromised given manager) and c_j (manager re-compromised given worker) of the same run, under independence across workers. $w = 2$ for all three loops.

model	$\sum_j a_j c_j$	rounds	predicted (last)	observed (last)	gap
Llama 3.3 70B	1.800	5	0.989	0.700	+0.289
DeepSeek V3.2	1.018	8	0.327	0.567	−0.239
qwen2.5:7b	0.578	5	0.048	0.650	−0.602

The gaps are far outside sampling error (at $n = 40$ – 60 the standard error of an observed rate is ≈ 0.06 – 0.08), and their *sign* differs by model, so the discrepancy is not a uniform bias that a single correction would absorb. The diagnostic is visible in the round-by-round data: after Llama’s manager falls to 0.725, its workers remain at 0.925 and 0.975 where the memoryless chain puts them at the manager’s own rate, and on DeepSeek the workers are compromised *more* often inside the loop (0.767) than their isolated per-edge rate (0.650) implies. Both patterns say that a binary per-round state is not sufficient here either—the same conclusion the content-form ablation reaches for the chain—so we report the loop as a directional mean-field prediction and state explicitly that the finite-chain

formulation, as written, is falsified by our measurements rather than supported by them.

Two caveats. First, ρ is computed from *isolated* survivals that Sections 5.2–5.7 of the main paper show to be unreliable; the Llama and qwen verdicts survive because neither $\sum_j a_j c_j$ is marginal (1.80, 0.58), whereas DeepSeek’s 1.018 sits close enough to the critical value that its supercritical label should be read together with its measured dynamics (endemic over eight rounds) rather than from the point estimate alone. Second, we study a single recurrent pattern at one temperature; the three models span five to eight rounds, enough to separate decay from saturation, not to demonstrate extinction.

Appendix G. Judge Calibration, Scope, and Known Failure Modes

What was measured. The compromise rule of Section 3.4 of the main paper ($C = 1[\text{ASV} \geq 0.9]$, ASV = fraction of target bigrams present) was calibrated on **ground-truth-labelled** outputs from the local model qwen2.5:7b ($n = 12$ per compromised class, $n = 12$ benign, $n = 3$ refusing outputs; exact-string injection, benign “verification code” framing). The labels are assigned by construction, not by a judge: *benign* outputs never see an injected instruction; *refusing* outputs see one and do not reproduce the target; *propagating* outputs are retained only when the target appears, and *direct* outputs come from the injected instruction alone.

Table 7: Calibration of the two continuous scores on labelled qwen2.5:7b outputs (mean $\pm$ sd). ASV separates the two compromised classes (1.000) from both benign classes (0.000) with no overlap; the full-containment variant instead scores benign outputs as high as 0.459, which is what motivated the switch (reported in the main paper).

score	benign	refusing	propagating	direct
ASV (target-bigram containment)	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
MR (Dice vs. direct output)	0.253 $\pm$ 0.179	0.484 $\pm$ 0.044	0.598 $\pm$ 0.070	0.862 $\pm$ 0.212
full-containment (rejected)	0.459 $\pm$ 0.215	0.597 $\pm$ 0.274	0.729 $\pm$ 0.223	0.825 $\pm$ 0.274

Agreement at the deployed threshold. At $\tau_{\text{ASV}} = 0.9$ the rule attains precision 1.000, recall 1.000, F1 1.000, FPR 0.000 and FNR 0.000 on this set (TP = 24, FP = 0, TN = 15, FN = 0); the closest benign score observed anywhere in the probes is 0.875, i.e. the threshold sits inside a gap rather than on a boundary. The Dice-based MR branch, by contrast, separates the same classes at accuracy 0.949 with FPR = 0 only for $\tau_{\text{MR}} > 0.53$, and the rejected full-containment variant reaches accuracy 0.795 with FPR = 0.067 at its own optimal cut—consistent with the main paper’s 1.000 vs. 0.795 comparison.

Scope of this validation, and what it does not establish. The calibration covers **one model** (qwen2.5:7b), **one payload family** (exact-string targets), and **no defence**: both the benign and the compromised classes are drawn from undefended runs. The rule is deterministic and needs no per-model recalibration, but we do *not* claim that the same operating point transfers unchanged to frontier models at temperature 0.7, to paraphrase- or obfuscation-obfuscated targets, or to defended runs. Two further failure modes follow from the operational definition and should be read alongside

every number we report: (i) a paraphrased leak that destroys the target’s bigrams is scored as *not* propagating (false negative), and (ii) a safe quotation of the target inside a benign summary is scored as propagating (false positive). Validation on frontier models, multiple payloads and defended conditions requires human annotation of a stratified subset across those conditions; we report the protocol for it below so the limitation is precise rather than open-ended. Read against a human annotator, both failure modes concern the *semantics* of the target: a paraphrase that destroys its bigrams is scored as not propagating, and a safe quotation inside a benign summary is scored as propagating.

Outcome levels. Every judged activation is reported at level (a) *textual propagation*—the target string/code fragments appear. The calibration set contains no case that separates (a) from (b) *instruction adoption/compliance* within the same output, and no case of (c) *harmful action* (the payload is a benign code, so (c) is by construction outside this study). Adoption is a subset of propagation in this task family: an agent that acts on the injected instruction must surface the target, whereas the converse does not hold. Since every label here is positive on (a) whenever it is positive on (b), the judge is an *upper bound* on the adoption rate and we report (a) as the propagation outcome. We do not use it as evidence of instruction adoption or of harmful action anywhere in the paper.

Replay procedure. In the same spirit we state precisely what the content-form replay of Section 5.4 of the main paper does. Each replay trial draws a fresh receiver, rotates the benign assignment over the same pool, and consumes either content captured from a real natural run, a canonical artefact produced on demand, a fixed hand-written string, or the in-context content under a different sender label; temperature, token budget, system prompt, role, judge and thresholds are identical across arms, so the arms differ only in message form and in the label of its sender. The in-context arm is the harness self-check and has a target it must hit, namely the rate measured in-chain; the other arms are read as comparisons against it. Because each trial resamples the source output and draws the receiver fresh, a replay rate can differ from an in-chain rate through sampling noise and per-context variation as well as through message form. On the edge where the self-check misses (0.733 versus 0.947) we therefore report the value for completeness and do not use it as an estimate of the form effect. The local re-run of this harness (qwen2.5:7b, 4 natural trials and 2 replays per arm and edge) is consistent with that reading rather than with a validated harness: every edge sits inside the replay confidence interval, yet two of three edges differ from the in-chain rate by 0.500, which at this sample size the interval cannot resolve. We read the local run as *underpowered*, not as a passed self-check, and we do not report it as evidence about form; the $n = 30$ –40 configuration behind the content-form table of the main paper (its Table 4) remains the only configuration in which the self-check discriminates.

Behaviour on replayed activations (local model). Table 8 adjudicates every activation of a 24-item stratified subset drawn from the replayed harness of Section 5.4 of the main paper (qwen2.5:7b, one payload, no defence; the subset is stratified over arm and over judge verdict so that both classes are represented). Each output was read and labelled at the three outcome levels before the judge score

was consulted. Against level (a) the rule reproduces the labels exactly: precision 1.000, recall 1.000, $F_1 = 1.000$, $FPR = FNR = 0.000$. Against level (b) it is not a usable detector: it flags 19 activations, and none of them involves instruction adoption—every positive is the target quoted inside a work product (an executive summary, a ticket classification, a release note, a review annotation). Three activations that sit downstream of content that *did* carry the target do not carry it themselves, and the rule correctly reports them as negative; a reader who stopped at the upstream context would have scored them as propagating. We therefore report (a) and never (b), which is precisely the distinction the main paper draws when it calls the judge an upper bound on adoption.

Table 8: Adjudication of the 24-activation subset against the three outcome levels (level (a) textual propagation, (b) instruction adoption, (c) harmful action). Labels were assigned by reading the full outputs, before the judge score was consulted, with the help of an automated string-matching assistant and with review by one of the authors; the labelling was not independent of the paper’s authors, so these numbers are a consistency check on the rule, not an independent validation of it.

level	TP	FP	TN	FN	prec.	recall	FPR
(a) textual propagation	19	0	5	0	1.000	1.000	0.000
(b) instruction adoption	0	19	5	0	—	—	0.792
(c) harmful action	0	19	5	0	—	—	0.792

Protocol for a full validation (not yet executed). A stratified subset should be drawn to distribute evenly across no-defence/defence, judge success/failure, topology, model and arm; each case should be labelled at levels (a), (b) and (c) by a human annotator, blind to the judge score, with a second annotator on a subset for agreement. Reported metrics should be precision, recall, F1, FPR, FNR and the full 2×2 confusion matrix per level, with Wilson/Clopper-Pearson intervals, cluster bootstrap for intervals (clusters = payload $\times$ defence $\times$ arm $\times$ edge, since trials repeat within contexts), and worked TP/FP/TN/FN examples. The scripts that generate the stratified sheet and score it are part of the released artifact; the numbers in Table 7 are the portion we had measured at submission time.

Appendix H. Utility under Attack, and Estimator Validation

Table 9 pairs the propagation outcome with its cost. Redaction on the susceptible model blocks propagation at *no* utility cost, while paraphrase buys no benefit there and costs utility; on the resistant model both defences are pure cost, since there is nothing to block. Reporting (ASR, ΔU) jointly is what makes these statements checkable: judged on propagation alone, paraphrase and doing nothing are indistinguishable, and only the utility column exposes that paraphrase is strictly dominated.

Estimator validation. On synthetic data with known ground truth, Wilson intervals attain nominal coverage and our bootstrap test is unbiased ($|\text{bias}| \leq 0.10\sigma$, size 0.067–0.083 at $\alpha = 0.05$ over

Table 9: Utility under attack, 20 paired clean/attack runs. Retention = $U_{\text{attack}}/U_{\text{clean}}$; ΔU is the clean–attack gap.

model	defence	ASR	U_{att}	ΔU	ret.
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	0.400
DeepSeek	redact	0.000	1.000	+0.000	1.000
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

60 replications), with higher power than a CI-overlap rule (0.750 versus 0.625). Detecting a 0.10 deviation at power 0.8 needs $n \approx 200$; at $n = 40$ only deviations above 0.26 are excluded, so we treat no $n = 40$ null as evidence and label the one $ASR = \prod_i \hat{s}_i = 0$ cell as uninformative. At $n = 200$ the Llama failure is powered (MDE = 0.11, $\Delta = 0.615$, $p = 0.0002$) while DeepSeek remains a non-rejection at a tighter MDE 0.13 ($\Delta = 0.022$, $p = 0.60$); Nova, an $n = 40$ null ($p = 0.084$), instead rejects ($\Delta = 0.051$)—why we decline to read small- n nulls as consistency, though the deviation is small and sits on Nova’s floor. Trial-level assumptions were tested too: φ must be computed *within* a temperature, and a homogeneity test across the three benign task contexts rejects exchangeability on the weak edge ($\chi^2 = 13.40$, $p = 0.0014$), so our rates are context averages—the same context-dependence the content-form ablation exposes from the other direction.

Appendix I. Practical Implications, in Full

The four prescriptions of the Introduction change what is *reported* rather than what is built, and each addresses a specific measurement failure our results expose.

Composition is safe once transportability is checked. Because the chain identity is algebraic rather than a modelling choice, the product of in-chain per-hop rates always equals end-to-end ASR; the only step that can fail—and that single-agent benchmarks implicitly rely upon—is transporting isolated per-hop estimates into that product. A single long chain measured under both protocols is therefore a cheap diagnostic, and its *slope* is the reading: shallow (DeepSeek, +2.3 points per hop at $n = 10$) means isolated measurement is safe to compose, steep (qwen +11.8; Llama +6.8) means the in-chain regime must be measured directly. Any composition of isolated estimates should also state the graph assumptions it needs (no unmodelled shared ancestors or correlations that invalidate the product-of-complements form) and be supported by a transport check for the target graph class. The corollary for attack design is that re-packaging is cheap for the defender to lose and cheap for the attacker to use.

Defence evaluation requires headroom. A defence cannot remove more signal than its model permits: where the unprotected base rate is 0.10 (Claude Sonnet 4.5) redaction looks perfect because there is almost nothing to defend, while where it is 1.00 (Llama, susceptible cells) the same code is essential. Reporting a post-defence ASR without its no-defence baseline confounds the defence effect with the model’s floor, and the utility measurements add a second

dimension: semantic paraphrase is strictly dominated in every cell we measured (Appendix H).

Graph structure decides placement, not per-hop survival. In a feed-forward network the set of nodes that can affect end-to-end propagation is exactly the set on some entry–target path, and within that set every node is equally effective *only* when a single cut covers the graph—that is, when no two entry–target paths are disjoint. The practitioner’s question is therefore: compute the paths, and among the nodes they contain pick the one with lowest instrumentation cost. Per-hop survival answers only the prior question of whether deploying a defence is worthwhile at all; redundant paths are what promote placement to a genuine optimisation problem.

Worked example: an AutoGen/MetaGPT-style pipeline. R4 Q1 asks for a worked example in which the four prescriptions change an engineering decision that an aggregate ASR alone would not. Consider the “software-company” pattern of MetaGPT or an AutoGen group chat wired as planner → worker → reviewer → aggregator, with a content filter to be placed at one hand-off. A single end-to-end ASR distorts this decision in three concrete ways, all measurable from the numbers reported in the main paper.

Prescription 1 (do not compose isolated estimates): suppose the team benchmarks the worker role in isolation, as a single-agent injection benchmark would, and obtains the Llama weak-edge value $s_i^{\text{ctrl}} = 0.233$. Composing it with the other two edges gives $\prod_i s_i^{\text{ctrl}} = 0.233$, and the team concludes the attack dies early and needs no defence. The measured chain ASR is 0.850, because inside the network the same edge survives $s^{\text{nat}} = 0.875$ (a 3.75× underestimate); the depth-error slope would have flagged exactly this kind of non-transport before deployment.

Prescription 4 (placement is structural): on the chain every interior node lies on the single entry–target path, so a filter at the reviewer is as effective as at the worker (each brings predicted ASR to 0.000). On a fan-in star, however, a hardened branch node is useless while the centre (the node on every path) is decisive. The aggregate ASR cannot tell these apart; only paths and cut sets can.

Prescription 3 (R_0 is not a safety criterion): the seven-agent chain has $R_0 = 0.821$ and the star $R_0 = 0.420$, both “subcritical”, yet the target is reached in 0.450 and 0.725 of runs respectively. Relying on $R_0 < 1$ would licence skipping the filter on the star—the topology where the attack reaches the target most often.

The aggregate ASR answers “did this run fail?”; the per-hop transport check, the structural placement view and the no-defence baseline together answer “where, and where should we intervene?”—which is the question that changes the engineering decision.

Appendix J. Limitations and Threats to Validity, in Full

Utility is a contamination proxy, not a correctness oracle: the metric flags whether the injected target string appears in the final deliverable, not whether the legitimate task was answered correctly, and a task family with independently verifiable ground truth (arithmetic or code execution) would sharpen the trade-off and make paraphrase’s cost comparable across conditions.

Transportability failures are real but their prevalence is unknown. The clearest failure is Llama’s weak edge, powered to $\Delta = 0.615$, $p =$

0.0002, MDE = 0.11 at $n = 200$: isolation underestimates in-chain survival by 3.75×. Nova shows the same middle-edge signature and DeepSeek transports cleanly under the same test, so this covers three of our five models on one attack family and one chain depth; how often the failure occurs elsewhere is uncharacterised, and the depth-error slope is a diagnostic rather than a guarantee, since a short chain at low power can read as flat.

Protocol sensitivity is reported rather than hidden. Our initial significant composition failure on DeepSeek was an artefact of the fixed-artefact policy and vanished when artefacts were drawn fresh per trial ($p = 0.0030 \rightarrow p = 0.930$, Appendix A). The sensitivity flipped the sign of the deviation, so unstated protocol choices that alter a result are a class of threat that benchmark replication exercises routinely miss.

Scale, topology and defence coverage. Headline chain results span four to fifteen agents; star and tree cells are replicated across three models with two Llama instances ($n = 7$ and $n = 10$); the recurrent case covers three models with limited power. Configurations we do not vary include networks of tens of agents, mesh, debate or mixture-of-experts topologies, re-injection, and defence families beyond redaction and paraphrase—detection layers, hop isolation and structured queries are referenced but not evaluated. Cell sizes of 30–40 trials give MDE ≈ 0.26 , and the ASV threshold applies to a literal target that would need recalibration for paraphrase-obfuscated ones.

Appendix K. Depth Profile: Per-Model Table and Rank Statistics

Table 10: Shape of the depth profile: relative error of the isolated product, $|\prod_i s_i^{\text{ctrl}} - P(C_i=1)|/P(C_i=1)$, against hops from the entry. range is min–max over informative depths; ρ is Spearman’s rank correlation between depth and error, slope its OLS trend in percentage points per hop. On Llama the chain dies at depth 11, so the last four depths are undefined.

model	chain	range	mean	ρ	slope
qwen2.5:7b	$n = 7$	31–89%	66%	+1.00	+11.8
Llama 3.3 70B	$n = 15$	0–92%	66%	+0.77	+6.8
DeepSeek V3.2	$n = 8$	7–35%	21%	−0.07	+0.1
DeepSeek V3.2	$n = 10$	4–33%	15%	+0.65	+2.3

Uncertainty on the slopes. The main paper quotes a permutation test and a bootstrap interval for each slope; Table 11 gives them in full. The permutation test is two-sided on Spearman’s ρ : where the number of informative depths is at most nine we enumerate all $n!$ relabellings exactly, otherwise we use 10^4 Monte Carlo permutations (seed 7); the intervals are percentile bootstrap (10^4 resamples of the (depth, error) pairs). Two consequences deserve emphasis. First, a perfect rank correlation at these sample sizes is still only $p = 0.003$: the test is weak by construction, so “significant” here means “not explainable by chance at $n = 6$ depths”, not that the slope is precise. Second, the DeepSeek profiles do not support a trend at either sample size—at $n = 10$ the interval for the slope spans zero ($[-0.3, +4.0]$) and the rank test does not reject ($p = 0.067$), and

at $n = 8$ the estimate is indistinguishable from flat ($\rho = -0.07$, $p = 0.91$)—which is why the main paper reports the DeepSeek profile as shallow rather than as evidence of a positive trend. We therefore describe the slope as a diagnostic, not as a calibrated correction.

Table 11: Permutation test and bootstrap intervals for the depth slopes of Table 10. n is the number of informative depths; “dropped” counts depths excluded because the measured or isolated rate was zero and the relative error is undefined.

model	n	ρ [CI95]	p_{perm}	slope	CI95
qwen2.5:7b ($n=7$)	6	+1.00 [1.00, 1.00]	0.003	+11.8	[+7.8, +14.8]
Llama 70B ($n=15$)	10	+0.77 [0.30, 0.95]	0.014	+6.8	[+2.7, +11.4]
Llama 70B (re-run)	10	+0.76 [0.30, 0.94]	0.014	+6.3	[+2.0, +11.0]
DeepSeek V3.2 ($n=8$)	7	-0.07 [-0.89, 0.88]	0.906	+0.1	[-5.3, +4.5]
DeepSeek V3.2 ($n=10$)	9	+0.65 [-0.07, 0.98]	0.067	+2.3	[-0.3, +4.0]

The slopes form a spectrum rather than two categories, and the ordering is a consequence of what the transportability measurements show directly: where the isolated rate transports best (DeepSeek matches its measured ASR to 2.2%) composing edge estimates adds little systematic error, whereas an edge underestimated by 3.75× (Llama’s middle edge) enters every product containing it and compounds with depth. The tree cell adds a second constraint—the error’s *sign* is not fixed: there the isolated estimate is optimistic, predicting certain compromise ($\prod_i s_i^{\text{ctrl}} = 1.000$) against a measured 0.800, and the two edges on the target’s path are exactly those whose isolated survival exceeds their in-chain survival. The defensible statement is therefore that *the slope diagnoses transportability while the direction remains model- and topology-dependent*, and independence itself is not what fails: evaluated on in-chain rates, the percolation recurrence reproduces the measured ASR in all three topologies (Appendix E).

Appendix L. Ethical Considerations and Artifact Availability

All experiments use a *benign* marker payload against our own harness via standard provider APIs; the attack forms are already documented for single agents. No real user data, production deployments or third-party systems were involved, and no AI system is listed as an author.

Code, per-cell outputs, tests and offline re-analysis (Appendices A–K) reproduce every table with no model call, including on a free local model (qwen2.5:7b); REPRODUCE.md maps each table to the command that regenerates it. The one component that is data rather than computation—the annotated subset behind the judge adjudication of Table 8—is released as a CSV together with the scoring script that turns it into the reported confusion matrix.

Appendix M. Role Order Versus Position: a Permutation Control

In the main paper the four roles are assigned to the chain in a fixed order (planner, worker, reviewer, aggregator), so any statement about *role*-dependent survival is confounded with position: the reviewer edge is always the third edge. To separate the two we re-ran

the same chain with the role labels permuted, moving the reviewer to the head and the planner to the third position, and compared survival per edge. Model, payload, task family, temperature, judge and trial count were held fixed; only the role assignment changed.

Table 12: Role-order permutation control (qwen2.5:7b, five-agent chain, no defence, 40 natural runs per ordering, BANANA-77 marker). In the permuted ordering the reviewer moves to the head of the chain and the planner to the third position, so each role except the duplicated one sits at a different depth. Values are in-chain per-edge survival s^{nat}_i .

depth	role (default)	s^{nat}	role (permuted)	s^{nat}
1	worker	0.425	worker	0.525
2	reviewer	0.882	planner	0.381
3	aggregator	1.000	aggregator	1.000
4	planner	0.733	planner	0.875

Table 13: The two readings of the permutation, as mean absolute differences. Under the position reading the k -th deepest edge of the default ordering is compared with the k -th deepest edge of the permuted ordering; under the role reading each role’s survival is compared across the two orderings. The smaller value indicates which variable the profile follows.

reading	mean $ \Delta $
by position (4 pairs)	0.186
by role (3 pairs)	0.151

The two readings are close (0.186 versus 0.151, a difference of 0.035 on a $[0, 1]$ scale), and they are not directly comparable in count: the position reading uses every shared depth, whereas the role reading can only use roles that occur in both orderings, and the duplicated role at the tail of a five-agent cycle reduces it to three pairs. The informative quantity is therefore not which mean is smaller but the size of the displacement itself: permuting the labels moves the *depth profile* by 0.186 on average, with the second edge moving from 0.882 to 0.381. Two consequences follow, and the main paper is written to respect both. First, the profile is *not* invariant to the role labels, so the role assignment cannot be ignored. Second, a single permutation cannot separate role from position: with near-equal distances and three to four pairs, claiming that survival follows the role rather than the depth would over-read the control. We therefore treat the role profile of the main paper (its Fig. 2) as a profile under one fixed assignment (planner, worker, reviewer, aggregator) rather than as a role effect, and we do not use it to rank roles.

Two further limits. The control was run on a single local model: the same experiment on Llama 3.3 70B was the intended second half of the comparison, and the Bedrock credential available to us expired before it could be run, so the reviewer’s request to repeat this for both models is only half answered. And at $n = 40$ runs per ordering the per-edge standard error is roughly 0.07–0.08, comparable to the displacement we report, so a larger set of permutations would be needed to estimate a role effect even if one exists.

Appendix N. Disclosure of AI-Assisted Technologies

Tool and version. **DeepSeek V4.1 Flash**, used interactively through the DeepSeek Harness web interface as a multi-turn session rather than as single completions. No other AI system was used, and no AI system is listed as an author.

Where it was used. Routine and permitted: polishing and formatting of text, and implementing the accompanying Python package and its test suite; every citation was retrieved and read in the original by the authors, and no citation, venue or numerical claim was taken from a model summary. *Methodology, disclosed under the policy:* the tool contributed to (i) the validation apparatus added during revision—the stratified annotation subset, the three-level labelling of Table 8, the two-condition pass rule for the replay self-check, and the cluster-bootstrap scoring script; (ii) the per-edge transport tests and the diagnostics in Section 5, including the two-sided equivalence test, the power and multiplicity summary of Section 5.9, and the depth-slope check; and (iii) the separation of the outcome into the three levels of Section 3.4 and the decision to report only textual propagation. The measurement protocols themselves, the model runs and every reported number were produced by the authors; where the tool implemented or re-analysed a measurement, the authors verified the result against the stored logs before it was reported. The labels of Table 8 were assigned by reading full outputs with the aid of an automated string-matching step and reviewed by an author; they are *not* independent human annotation, and are reported as a consistency check rather than a validation.

Prompts. As the interaction was a dialogue rather than a single prompt, we quote the operative instructions as issued: “*audit the manuscript adversarially against its own code and data rather than against its claims*”; “*verify every asserted defect on the source before acting on it*”; “*prefer correcting a result over narrowing its assumptions*”; “*report what the measurements show even where they contradict the paper*”. Two proposed defects were withdrawn under the second instruction because the committed results did not support them, and the same check caught an annotation error of our own: three activations first labelled as propagating do not contain the target in the full output, and were corrected before reporting. The complete session log is available to the programme chairs on request.

Accountability. The authors take responsibility for the accuracy of this work. Every number reported in the paper and in this supplement is regenerated by the released code without model calls, with one exception that is data rather than computation: the annotated labels behind Table 8 are released as a file together with the scoring script that reproduces the reported confusion matrix. The limits of the contribution—that the transportability failure is characterised on three of five models at one chain depth, that the judge is validated on a single local model and measures textual propagation rather than adoption, and that one of our own early composition results was an artefact of the fixed-artefact policy—are stated in the paper and in Appendix J rather than left for a reader to discover.

Appendix O. Testing Eq. (3) Where It Can Fail: a Reconvergent DAG

On the chain, the balanced tree and the single-entry fan-in star, each node has effectively one entry–target path, so the percolation recurrence of Eq. (3) collapses to the chain identity of Eq. (1) and cannot be tested. We therefore ran the harness on a reconvergent DAG in which the two parents of the target share an upstream node, $a0 \rightarrow a1 \rightarrow \{a2, a3\} \rightarrow a4$, which is the case the recurrence is not guaranteed to satisfy when the parent-reach events are correlated. The run used the local model qwen2.5:7b at $T = 0.7$, 40 natural runs and 30 replays per arm and edge (600 judged activations, 240 of them at the target).

Table 14: Reconvergent DAG ($a0 \rightarrow a1 \rightarrow \{a2, a3\} \rightarrow a4$, qwen2.5:7b, 40 natural runs, 30 replays per arm and edge). Top: per-edge in-chain versus isolated survival. Bottom: the target’s compromise probability under the chain product, under the percolation recurrence from in-chain marginals, under the same recurrence from isolated marginals (what a single-agent benchmark would compute), and as measured.

edge	s^{nat} (in-chain)	s^{ctrl} (isolated)
$a0 \rightarrow a1$	0.375	0.400
$a1 \rightarrow a2$	0.933	0.933
$a1 \rightarrow a3$	0.800	0.367
$a2 \rightarrow a4$	1.000	0.533
$a3 \rightarrow a4$	1.000	0.933
target compromise probability		
$\prod_i s_i^{\text{nat}}$ (chain product; wrong structure here)		0.280
Eq. (3) from in-chain marginals		0.545
Eq. (3) from isolated marginals		0.309
measured on natural runs		0.792 [0.736, 0.838]

Two readings follow and must not be conflated. First, the recurrence built from the *in-chain* marginals underestimates the measured rate by 0.247, which is what one expects if the two parent-reach events are positively correlated; the harness’s own self-check passes at the target (0.717 replayed versus 0.792 measured, $\Delta = -0.075$), so the gap is not an artefact of a failed replay. Second, and more consequential for practice, the recurrence built from the *isolated* marginals gives 0.309 against a measured 0.792: a single-agent benchmark that measures each edge in isolation and composes the results underestimates reachability on this graph by roughly 2.6×, and the per-edge discrepancies are concentrated ($a1 \rightarrow a3$: 0.800 in-chain versus 0.367 isolated; $a2 \rightarrow a4$: 1.000 versus 0.533). The direction matches the Llama chain failure of the main paper: isolation *under*-estimates what a hand-off achieves once the payload arrives inside an agent’s work product. Two limits of this measurement: three of the five edges do not reproduce their in-chain rate under replay (differences of 0.14–0.30 at $n = 30$ per arm), so the in-chain marginals carry that uncertainty; and the result is one model on one small DAG, so we read it as evidence that the recurrence is not exact in general rather than as a calibrated correction.

Table 15: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge, fixed-artefact protocol throughout). The three topologies have similar mean per-hop survival but differ substantially in outcome. The star-tree ASR gap is within noise at this size ($p = 0.60$); the chain-tree and chain-star gaps survive the family-wise control of Appendix B, so the “most permissive” ordering is numerical only. R_0 is reported without an interval: its numerator and denominator are formed per agent-instance and are not recoverable from the per-cell files we archived, so the interval we would have to print would not be reproducible from the release.

topology	edges	$\bar{s}$	ASR	R_0
chain	6	0.761	0.450	0.821
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	0.800	0.831

Appendix P. Topology at Matched Agent Count

The star-tree relation reverses with scale rather than merely shrinking: at $n = 7$ the tree exceeds the star numerically (0.800 vs. 0.725; Appendix B shows the gap is within noise, Fisher $p = 0.60$), whereas repeating the comparison at $n = 10$ puts the star above the tree on Llama (0.675 vs. 0.650) and, at $n = 7$, on DeepSeek (0.750 vs. 0.475). Nova sits at the resistance floor in both (star 0.000, tree 0.125). The direction of the topology effect is thus itself scale-dependent, which is a further reason not to summarise a network with a single per-hop scalar.

Appendix Q. What the Artifact Contains, and What It Does Not

We separate what is actually supplied from what a reader would need in order to regenerate a cell from scratch, so that the reproducibility statement matches the submission rather than describing an ideal package.

Supplied. The released code reproduces every table and figure in the paper and in this supplement *without live model calls*, from per-cell result files that are included: per-cell summaries (`results.json`), raw hop and agent logs, the offline re-analysis scripts (threshold rescoring, multiplicity control, cluster-bootstrap intervals, per-edge transport tests including the TOST analysis of Appendix H), the estimator-validation battery on synthetic data, and the two files behind the judge adjudication of Appendix G (the annotation sheet and the scoring script). `REPRODUCE.md` maps each table to the command that regenerates it, and the local tier runs on a free model (qwen2.5:7b) with no API key.

Model identifiers, dates and decoding. The hosted models are addressed by inference-profile identifiers, and every per-cell file records the identifier it used, so the table below is read off the released results rather than from memory. Dates are the first and last collection day observed across the archived per-cell files. Decoding is fixed at $T = 0.7$ with a 400-token cap for every hosted cell in the paper, and the local tier runs qwen2.5:7b at the same settings through an OpenAI-compatible endpoint.

Table 16: Hosted and local models actually referenced by the archived cells, with the date range over which their cells were collected. Cells, not calls, are counted; the mock backend appears because one synthetic self-test cell is shipped with the bundle.

identifier as recorded	backend	cells	dates
us.meta.llama3-3-70b-instruct-v1:0	bedrock	14	09-11-09-14
deepseek.v3.2	bedrock	10	09-11-09-14
amazon.nova-pro-v1:0	bedrock	4	09-11-09-14
us.anthropic.claude-sonnet-4-5-20250929-v1:0	bedrock	4	09-11
qwen2.5:7b	local	3	09-11-10-08
mock (synthetic self-test, no model call)	—	1	09-11

Where the bundle is. The code, per-cell outputs, and manifest are submitted as supplementary material alongside this paper. The machine-readable inventory is at `artifacts/MANIFEST.json` (one row per cell) and the table-to-command map is at `REPRODUCE.md`. The code is released under MIT and the measurement data and per-cell outputs under CC-BY-4.0, matching the licence on the first page of the paper.

Not supplied, and therefore not claimed. In particular we do not supply: per-condition call counts and failure counts, wall-clock collection times per cell, or the full prompt templates for every arm. Our cross-model table also mixes artefact protocols, as its caption states, and the fresh-artefact replicate of the three rows that predate that policy is not part of the release. The protocol flag that distinguishes fixed from fresh artefacts is recorded in the *commands* listed in `REPRODUCE.md` rather than inside each per-cell file, so we label it as reconstructed provenance rather than as a field of the release. We therefore claim *partial* reproducibility: the numbers in this paper are regenerated by the supplied code from the supplied per-cell outputs, and the live calls that produced those outputs are reproducible only in the sense that the scripts and their configuration are provided.

Inventory. A machine-readable inventory accompanies the bundle with one row per cell: model, defence, payload, topology, context pool, protocol (fixed or fresh artefacts), number of natural runs, controlled trials per edge, evaluated n , and the script that produced it. Analyses are labelled *confirmatory* (the composition tests of Section 5.2 and the multiplicity correction of Appendix A) or *exploratory*; the exploratory analyses—topology, obfuscation, payload form, the recurrent loop and the depth profile—carry family-wise control within their own family at $q = 0.05$ (Appendix B), and every test stated in the paper is covered by that accounting rather than left uncorrected. The intervals we report use Wilson bounds, with a cluster bootstrap where trials repeat within contexts or runs.